Problem Set for 27 March 2025

Exercise 0 Explain (without proving) the *implicit expressions.*

1. $\ker g$ as a subspace of $\ker fg$.

> *Solution. The linear map*
>
> $$\ker g \to \ker fg, \quad [x \mapsto x].$$
>
> *It is well-defined and injective.*

2. $\mathrm{im}(fg)$ as a quotient space of $\mathrm{im}(g)$.

> *Solution.*
>
> $$? \to ?, \quad ? \mapsto ?$$

3. $\mathrm{Hom}(V, W) \times \mathrm{Hom}(U, V) \to \mathrm{Hom}(U, W)$ the bi-linear map of composition

> *Solution. The bi-linear map*
>
> $$\mathrm{Hom}(V, W) \times \mathrm{Hom}(U, V) \to \mathrm{Hom}(U, W), \quad (g, f) \mapsto g \circ f.$$

4. $\mathrm{Hom}(U, V) \to \mathrm{Hom}(U, W)$ induced by $f : V \to W$.

> *Solution.*
>
> $$? \to ?, \quad ? \mapsto ?$$

5. $\mathrm{Hom}(W, U) \to \mathrm{Hom}(V, U)$ induced by $f : V \to W$.

> *Solution.*
>
> $$? \to ?, \quad ? \mapsto ?$$

6. $\mathrm{Hom}_{\mathrm{Sets}}(S, V) = V^n$ for finite set $|S| = n$. Here $V^n = V \times \cdots \times V$, the Cartesian product or direct sum of n V's.

> *Solution.*
>
> $$? \to ?, \quad ? \mapsto ?$$

7. $\mathbb{F}[x]$ as a subspace of $\mathbf{Hom}_{\mathbb{F}}(\mathbb{F}[\![x]\!], \mathbb{F})$, as mentioned in `Ex_17-Mar-2025 (solution)`

> *Solution.*
>
> $$? \to ?, \quad ? \mapsto ?$$

8. The $\mathbb{C}$-linear space $\mathbf{Hom}_{\mathbb{R}}(\mathbb{C}, V)$, provided an $\mathbb{R}$-linear space V.

> *Solution. Figure out the definition of $\mathbb{C}$-linear combination*
>
> $$f + z \cdot g, \quad \forall f, g \in \mathbf{Hom}_{\mathbb{R}}(\mathbb{C}, V), \quad \forall z \in \mathbb{C}.$$

9. (**Optional**) The set of all subsets of X forms an $\mathbb{F}_2$-linear space, where $\mathbb{F}_2$ is a field with 2 elements.

> *Solution.*
>
> $$? \to ?, \quad ? \mapsto ?$$

Example Construct a *satisfying* linear isomorphism

$$\mathrm{Hom}_{\mathbb{F}}(U \times V, W) \simeq \mathrm{Hom}_{\mathbb{F}}(U, W) \times \mathrm{Hom}_{\mathbb{F}}(V, W).$$

Proof.

Step 1 Write down explicitly the linear map

$$\begin{array}{rccc} \Phi: & \mathrm{Hom}_{\mathbb{F}}(U, W) \times \mathrm{Hom}_{\mathbb{F}}(V, W) & \to & \mathrm{Hom}_{\mathbb{F}}(U \times V, W) \\ & (f, g) & \mapsto & [(u, v) \mapsto f(u) + g(v)]. \end{array}$$

For a very detailed expression,

$$\begin{array}{rccc} \Phi: & \mathrm{Hom}_{\mathbb{F}}(U, W) \times \mathrm{Hom}_{\mathbb{F}}(V, W) & \to & \mathrm{Hom}_{\mathbb{F}}(U \times V, W) \\ & \left(\begin{bmatrix} U & \xrightarrow{f} & W \\ u & \mapsto & f(u) \end{bmatrix}, \begin{bmatrix} V & \xrightarrow{g} & W \\ v & \mapsto & g(v) \end{bmatrix}\right) & \mapsto & \begin{bmatrix} U \times V & \xrightarrow{\Phi(f,g)} & W \\ (u, v) & \mapsto & f(u) + g(v) \end{bmatrix}. \end{array}$$

Step 2 Check that Φ is a well-defined linear map, i.e.

$$\Phi((f, g) + \lambda(f', g')) = \Phi(f, g) + \lambda\Phi(f', g'),$$

for both sides map (u, v) to $f(u) + f'(u) + \lambda g(v) + \lambda g'(v)$.

Step 3 Check that Φ is an injection. If $\Phi(f, g) = 0$, then so are the compositions

$$U \xrightarrow{\text{inclusion}} U \times V \xrightarrow{\Phi(f,g)} W, \quad u \mapsto (u, 0) \mapsto f(u),$$

and

$$V \xrightarrow{\text{inclusion}} U \times V \xrightarrow{\Phi(f,g)} W, \quad v \mapsto (0, v) \mapsto g(v).$$

It yields that $f = 0$ and $g = 0$.

Step 4 Check that Φ is an surjection. For any $\alpha : U \times V \to W$, consider

$$f_\alpha : U \to W, \quad u \mapsto \alpha(u, 0),$$

and

$$g_\alpha : V \to W, \quad v \mapsto \alpha(0, v).$$

Now $\Phi(f_\alpha, g_\alpha) = \alpha$, yielding that (f_α, g_α) is a preimage.

Step 5 Since Φ is well-defined, injective, and surjective, Φ is an linear isomorphism.

Your proof must include Step1-5.

Exercise 1 (optional if you have done this before) Show that $U \simeq \mathrm{Hom}_{\mathbb{F}}(\mathbb{F}, U)$ for any $\mathbb{F}$-linear space U.

> ***Step 1*** *Write down explicitly the linear map $u \mapsto [1_{\mathbb{F}} \mapsto u]$, to be explicit,*
>
> $$\Phi: \quad \text{Big Formula...}$$
>
> ***Step 2*** *Check that Φ is a well-defined linear map, i.e.,*
>
> ***Step 3*** *Show that Φ is injective, your solution...*
>
> ***Step 4*** *Show that Φ is surjective, your solution...*
>
> ***Step 5*** *Conclude that Φ is an linear isomorphism.*

Exercise 2 (optional if you have done this before) Show that $U \simeq \mathrm{Hom}_{\mathbb{F}}(\mathrm{Hom}_{\mathbb{F}}(U, \mathbb{F}), \mathbb{F}))$ if $\dim U < \infty$.

> ***Step 1*** *Write down explicitly the linear map* $u \mapsto [f \mapsto f(u)]$*, to be explicit,*
>
> $$\Phi : \quad \text{Big Formula...}$$
>
> ***Step 2*** *Check that* Φ *is a well-defined linear map, i.e.,*
>
> ***Step 3*** *Show that* Φ *is injective, your solution...*
>
> ***Step 4*** *Show that* $\dim U = \dim \mathrm{Hom}_{\mathbb{F}}(U, \mathrm{Hom}_{\mathbb{F}}(U, \mathbb{F}))$ *by induction.*
>
> ***Step 5*** *Conclude that* Φ *is an linear isomorphism by dimensional analysis.*

Exercise 3 Let V be a linear space and $S \subset V$ is linearly independent (S is not necessary finite). Show that

$$\mathrm{Hom}_{\mathbb{F}}(\mathrm{Span}(S), \mathbb{F}) \simeq \mathrm{Hom}_{\mathrm{Sets}}(S, \mathbb{F}).$$

> ***Step 1*** *Write down explicitly the linear map $f \mapsto (f(s))_{s \in S}$, to be explicit,*
>
> $$\Phi : \quad \text{Big Formula...}$$
>
> ***Step 2*** *Check that Φ is a well-defined linear map, i.e.,*
>
> ***Step 3*** *Show that Φ is injective, your solution...*
>
> ***Step 4*** *Show that Φ is surjective, your solution...*
>
> ***Step 5*** *Conclude that Φ is an linear isomorphism.*
>
> *In particular, $\mathbb{F}[\![x]\!] \simeq \mathrm{Hom}_{\mathbb{F}}(\mathbb{F}[x], \mathbb{F})$.*

Exercise 4 Recall that $\mathbb{C}$-linear spaces are automatically $\mathbb{R}$-linear, and so are $\mathbb{C}$-linear maps. Show that

$$\mathrm{Hom}_{\mathbb{R}}(U,V) \simeq (\mathrm{Hom}_{\mathbb{C}}(U,V))^2 \quad \text{(as } \mathbb{R}\text{-linear spaces).}$$

Here $W^2 := W \times W$.

> ***Step 0*** *Show that $\mathrm{Hom}_{\mathbb{C}}(U,V)$ is indeed an $\mathbb{R}$-linear space.*
>
> ***Step 1*** *Write down explicitly the linear map $f \mapsto (f_1, f_2)$, to be explicit,*
>
> $$\Phi: \quad \text{Big Formula...}$$
>
> - *Hint: show that $f_1(u) := f(u) - if(iu)$ is $\mathbb{C}$-linear, and so is $f_2 := (\overline{f})_1$.*
> - *Warning: for set-theoretic issues, there are no natural ways to define conjugation for arbitrary complex vector spaces. You can simply take $\dim U, \dim V < \infty$ for convenience.*
>
> ***Step 2*** *Check that Φ is a well-defined linear map, i.e.,*
>
> ***Step 3*** *Show that Φ is injective, your solution...*
>
> ***Step 4*** *Show that Φ is surjective, your solution...*
>
> ***Step 5*** *Conclude that Φ is an linear isomorphism.*
>
> ***(Optional Exercise)*** *Determine*
>
> $$\mathrm{Hom}_{\mathbb{R}}(U,V) \overset{?}{\simeq} (\mathrm{Hom}_{\mathbb{C}}(U,V))^2 \quad \text{(as } {\color{red}\mathbb{C}}\text{-linear spaces).}$$
>
> *Hint: There are at least 2 different $\mathbb{C}$-linear structures over $\mathrm{Hom}_{\mathbb{R}}(U,V)$, e.g.,*
>
> 1. $(z \underset{1}{\cdot} f) : U \to V, \quad u \mapsto f(z \cdot u)$;
> 2. $(z \underset{2}{\cdot} f) : U \to V, \quad u \mapsto z \cdot f(u)$.

Problem (optional) Show that V is a subspace of some $\prod_{x \in X} \mathbb{F}$, **if and only if** the canonical map

$$V \to \mathrm{Hom}_{\mathbb{F}}(\mathrm{Hom}_{\mathbb{F}}(V, \mathbb{F}), \mathbb{F}), \quad v \mapsto [f \mapsto f(v)]$$

is injective. The answer is already shown in `supplementary reading material`.